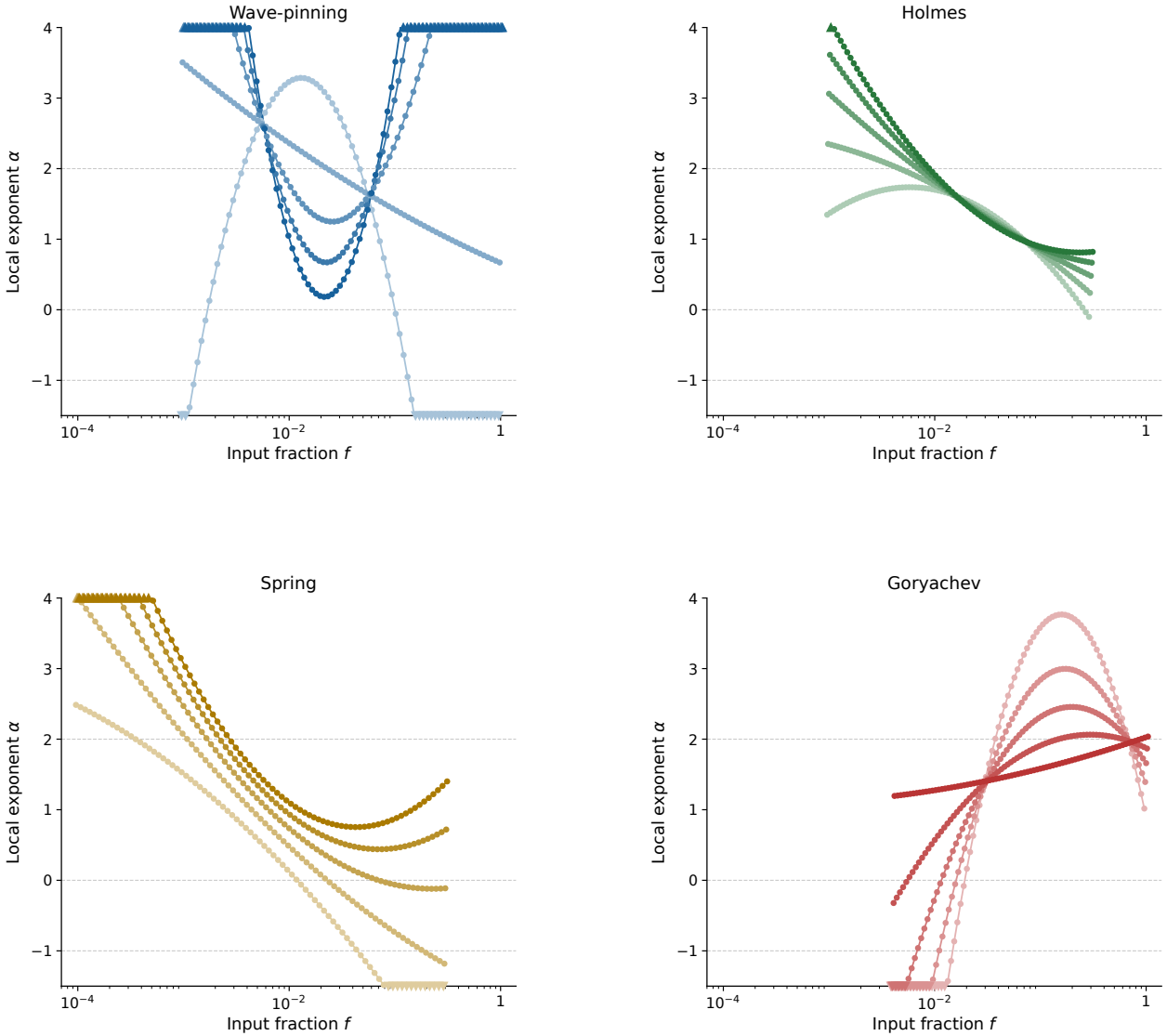
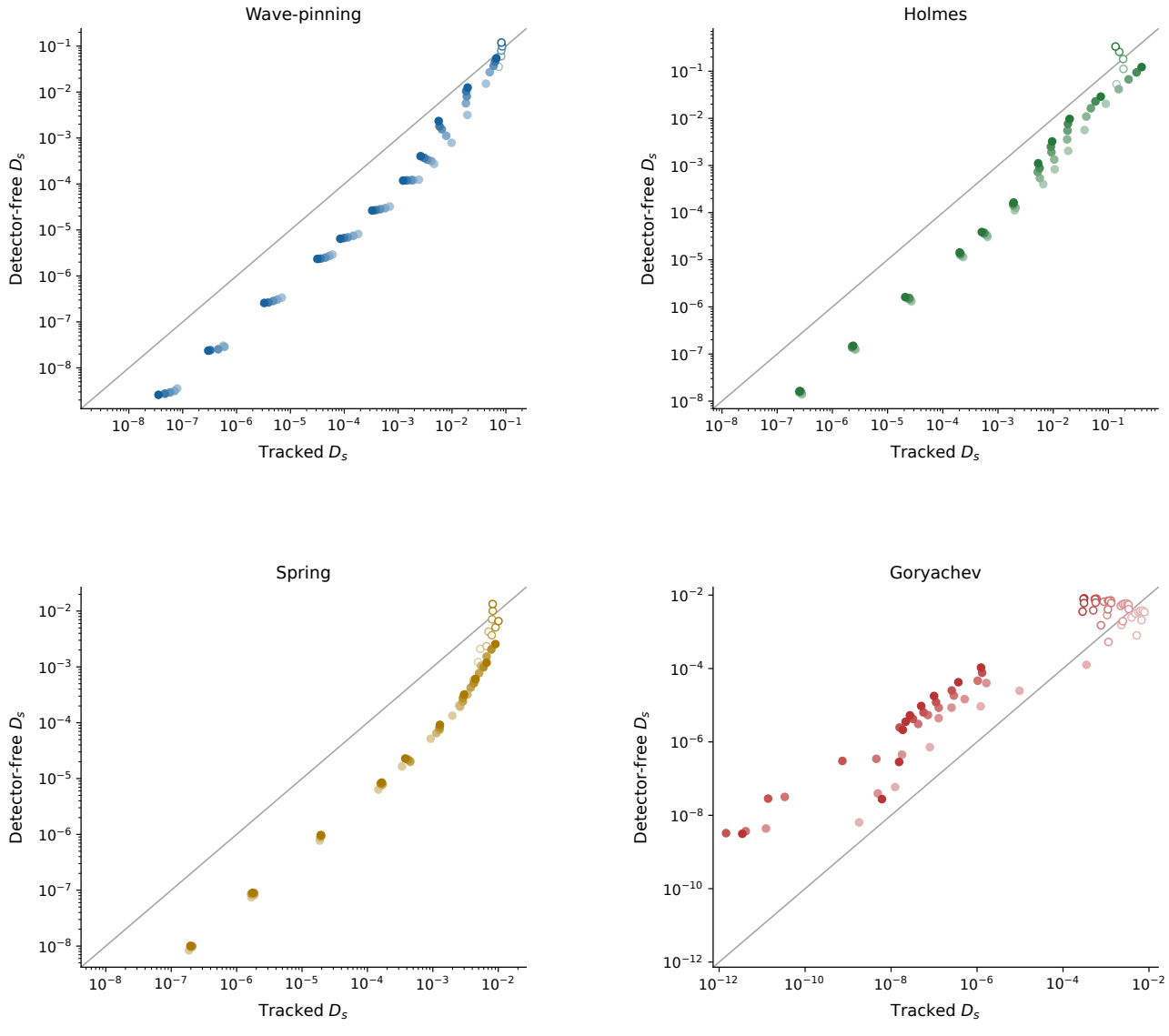
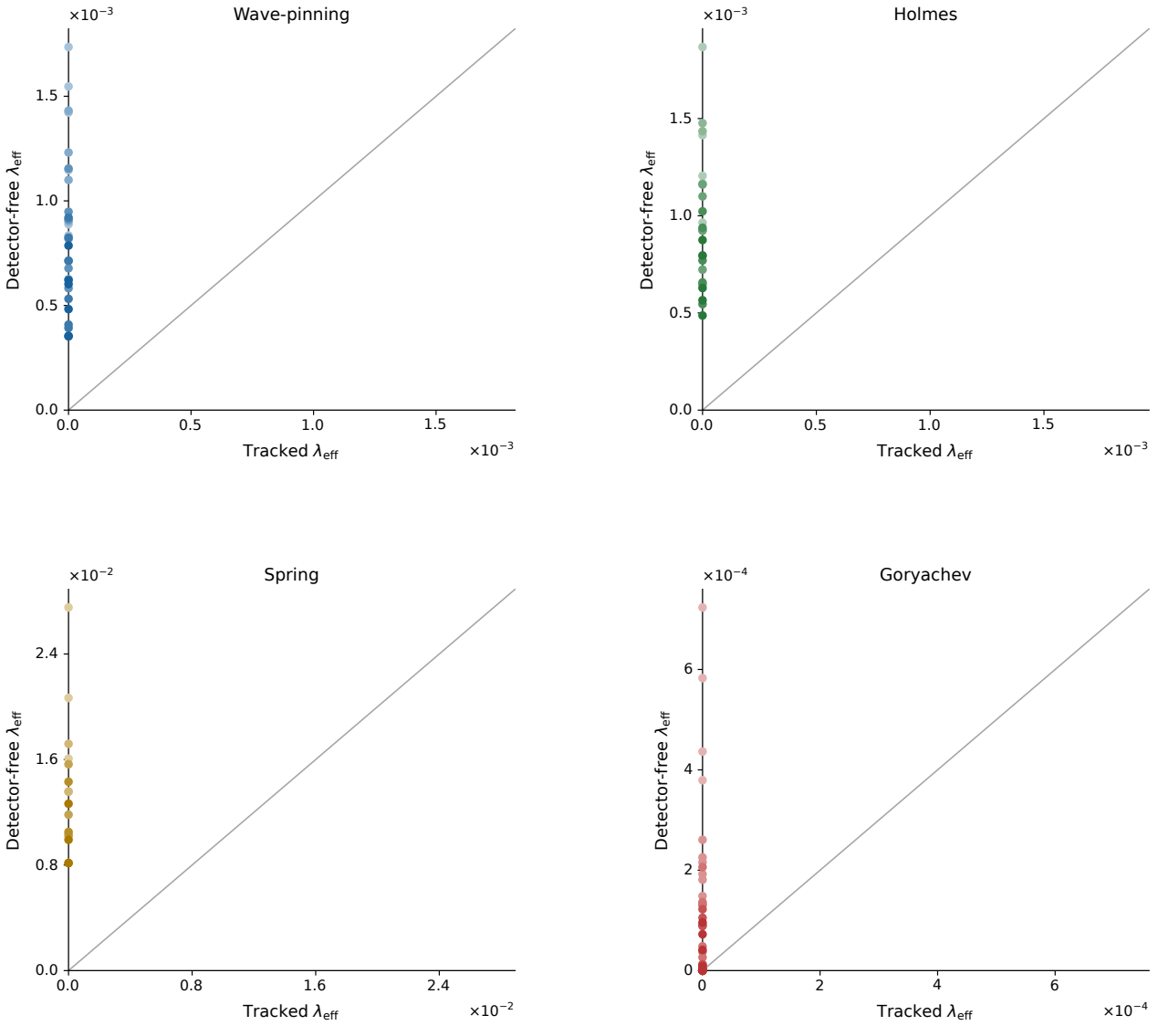
Smooth diagnostic size–input surface

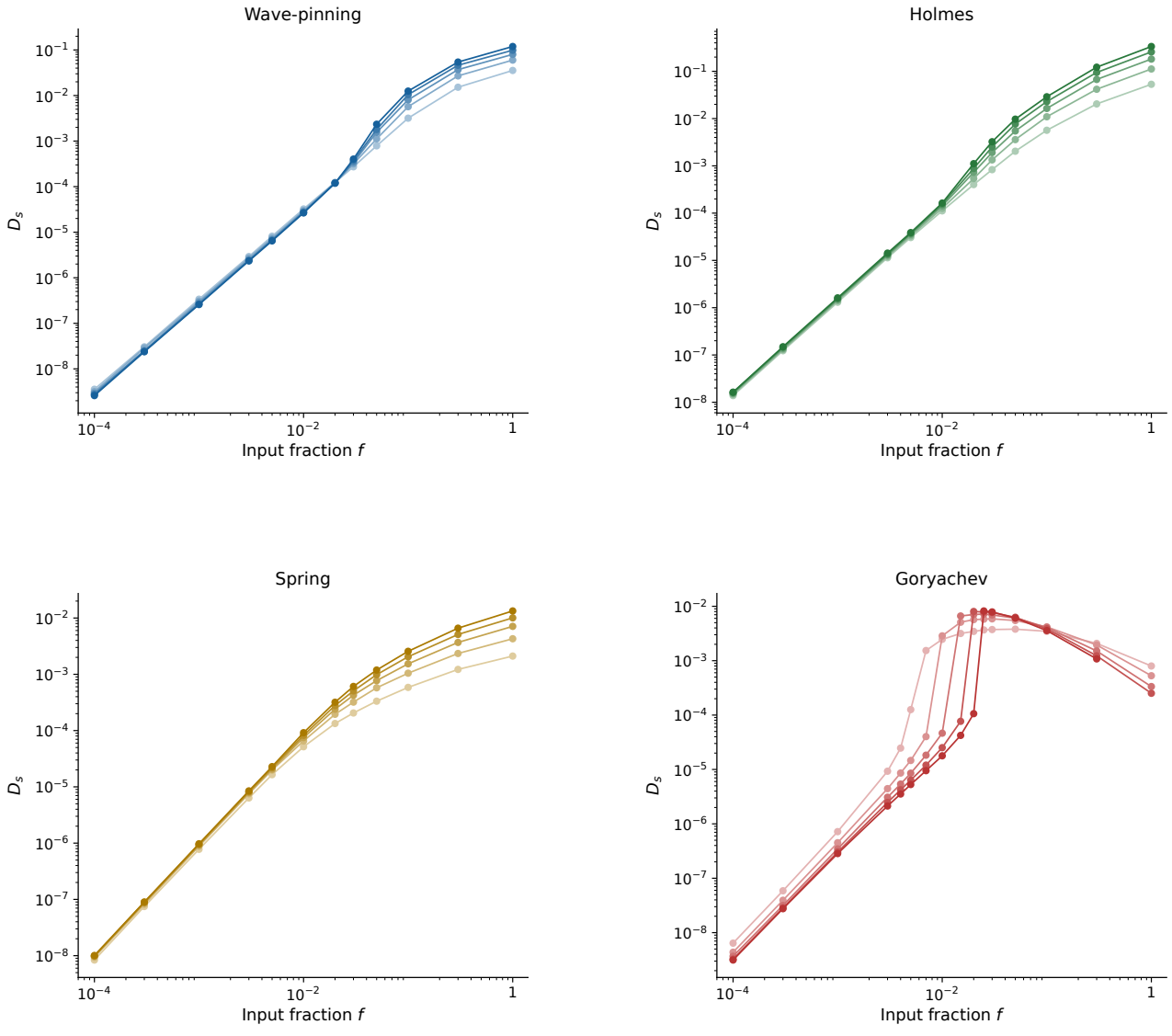


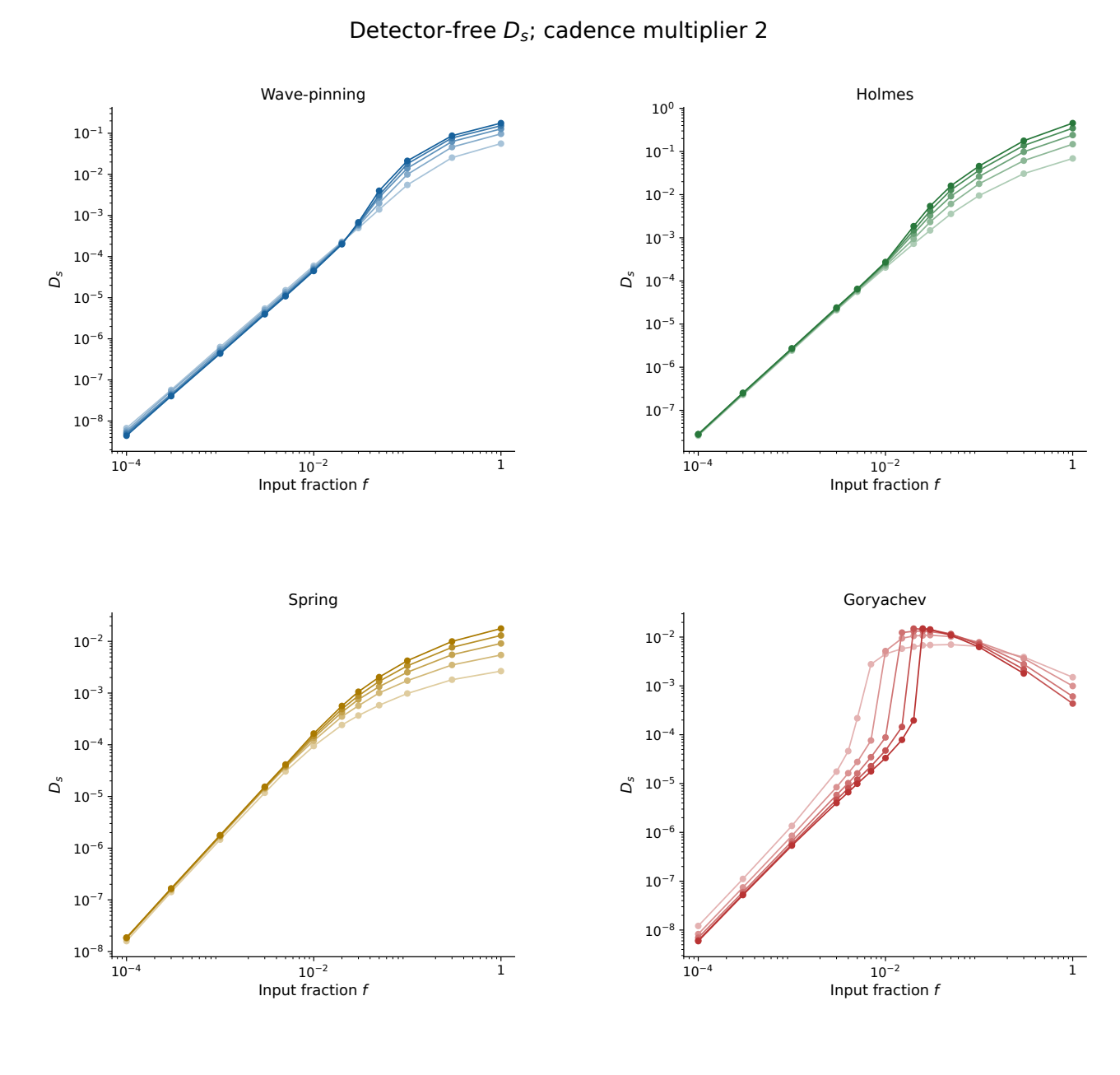
Fixed degree-two tensor polynomial in ln k versus ln circumference and ln input; diagnostic only. No missing local estimate is filled in the main figure.

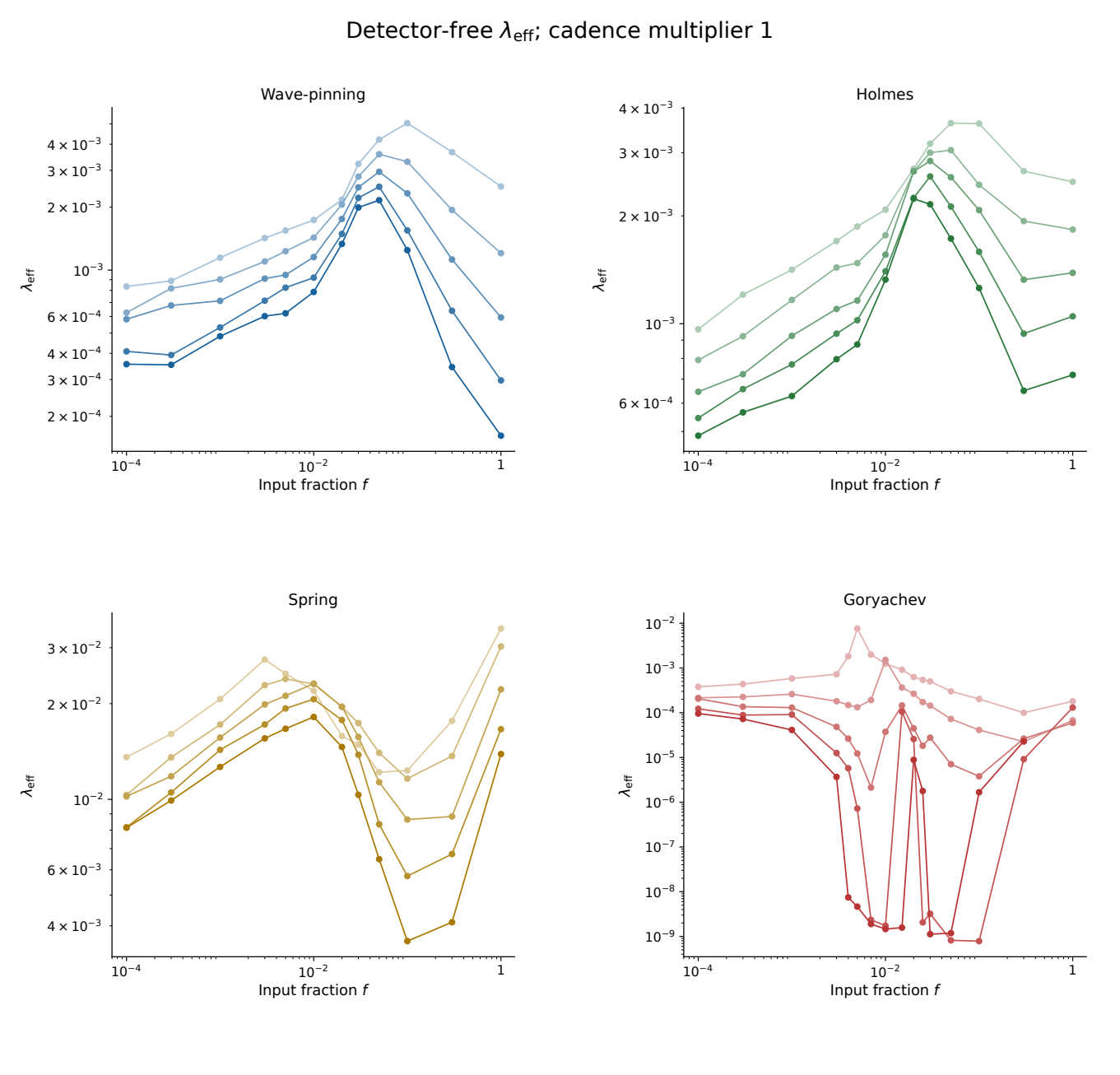

Detector-free versus tracked D_s; five circumference shades
Wave-pinning
Holmes
Spring
Goryachev
Detector-free D_s
Tracked D_s

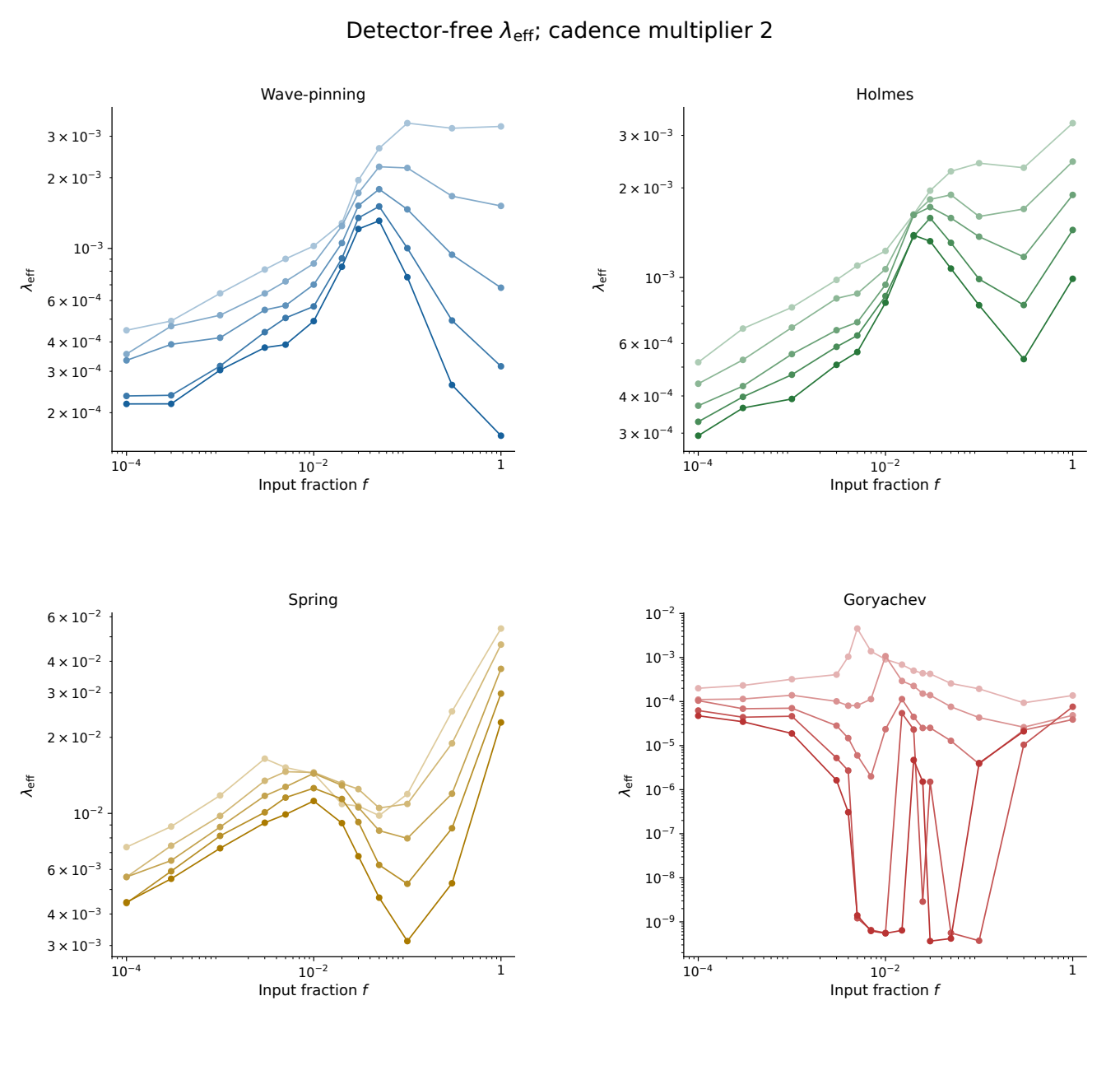


Linear axes retain empirical zeros; only undefined pairs are omitted. Open markers identify forced conditions; no undefined value is imputed.

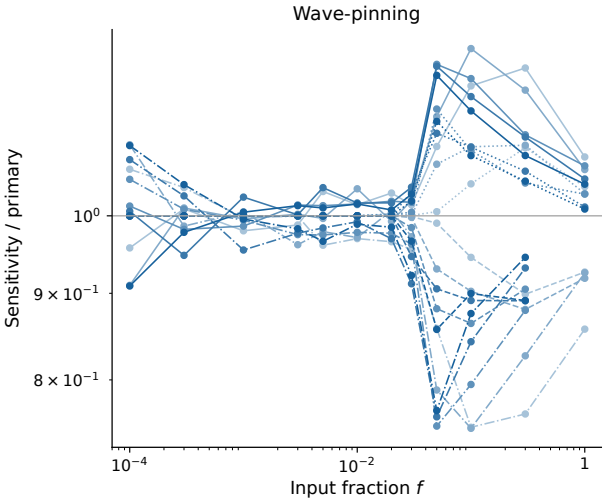

Detector-free D_s; cadence multiplier 1
Wave-pinning
Holmes
Spring
Goryachev
D_s
Input fraction f
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-4}
10^{-2}
1

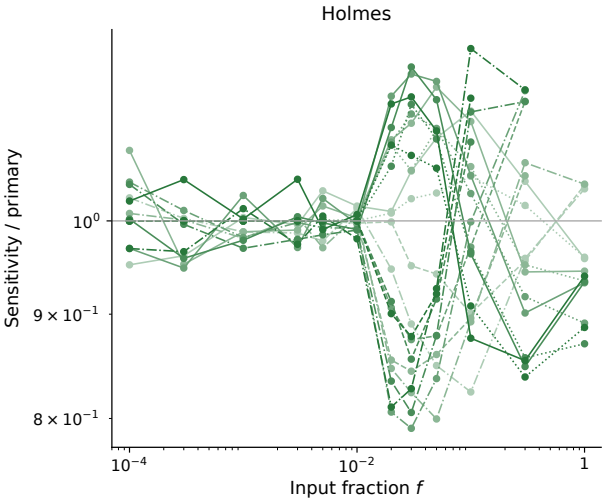

Detector-free D_s; cadence multiplier 2
Wave-pinning
Holmes
Spring
Goryachev
D_s
Input fraction f

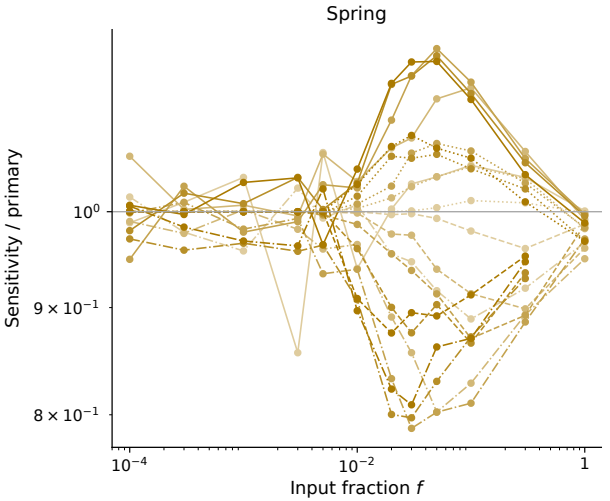

Detector-free λ_{eff}; cadence multiplier 1
Wave-pinning
Holmes
Spring
Goryachev
λ_{eff}
Input fraction f

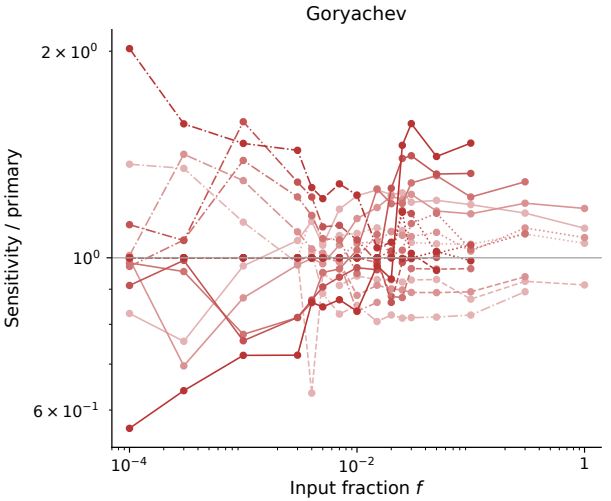

Detector-free λ_{eff}; cadence multiplier 2
Wave-pinning
Holmes
Spring
Goryachev
λ_{eff}
Input fraction f

Tracked sensitivity: D_s

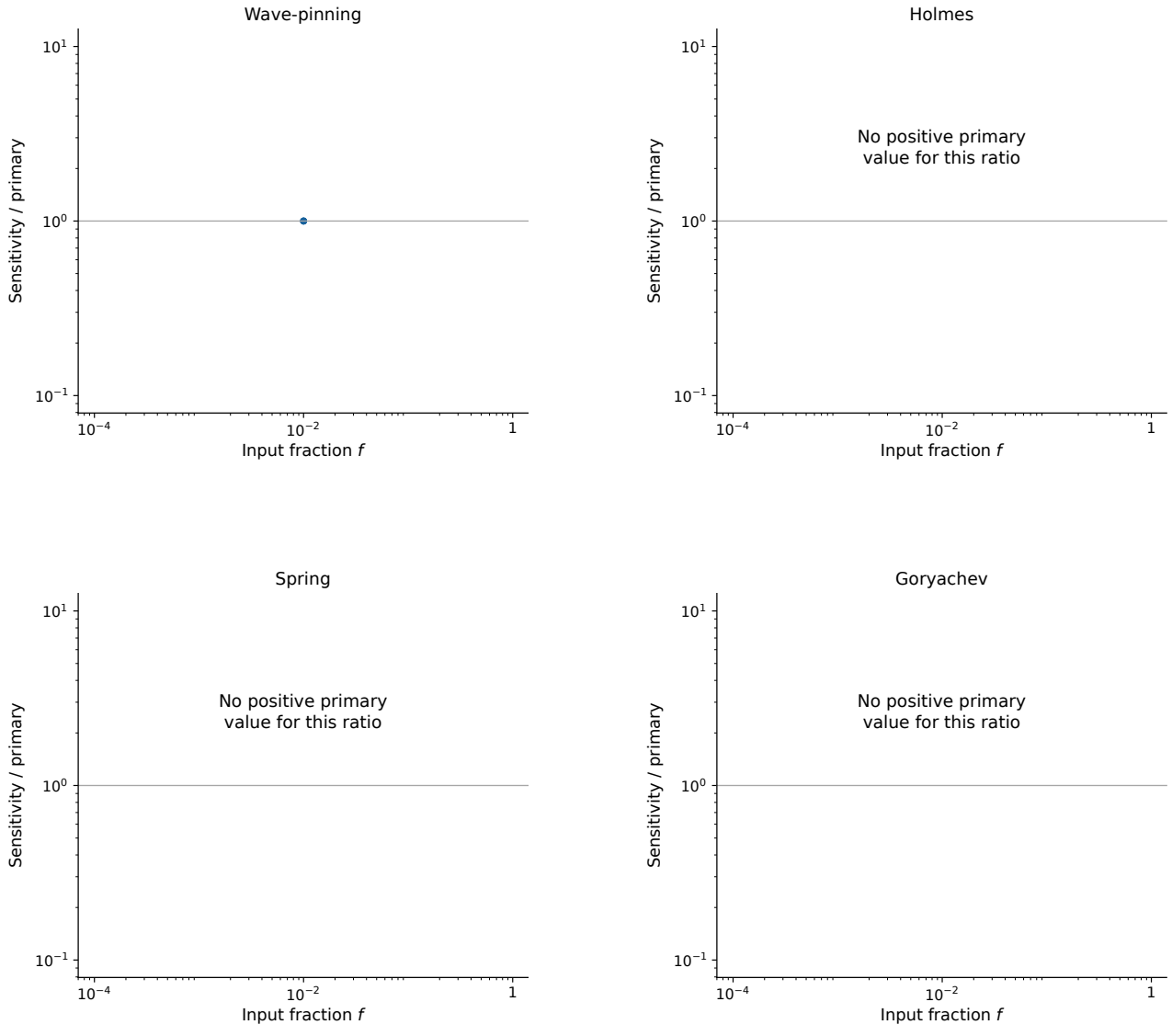




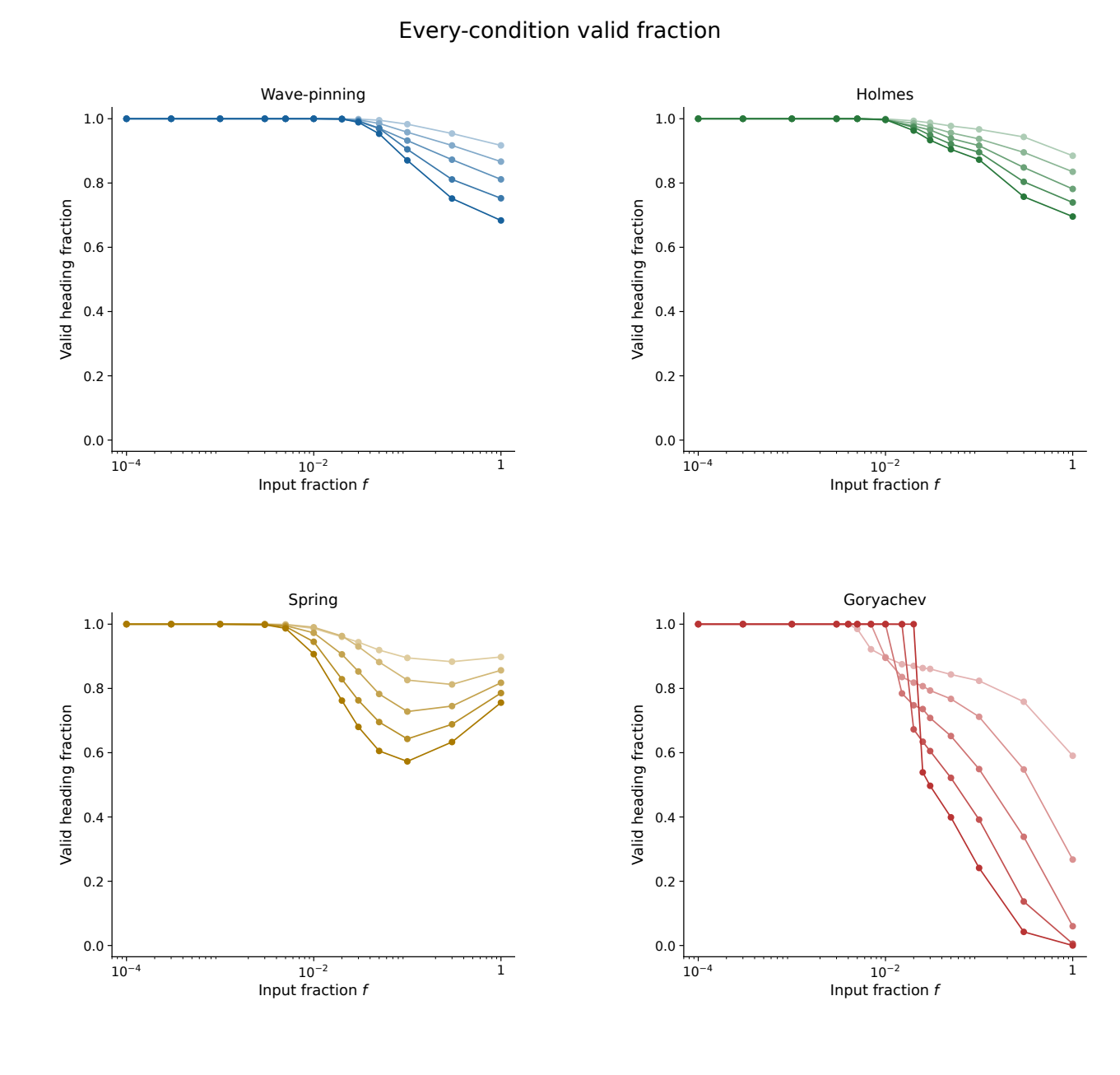


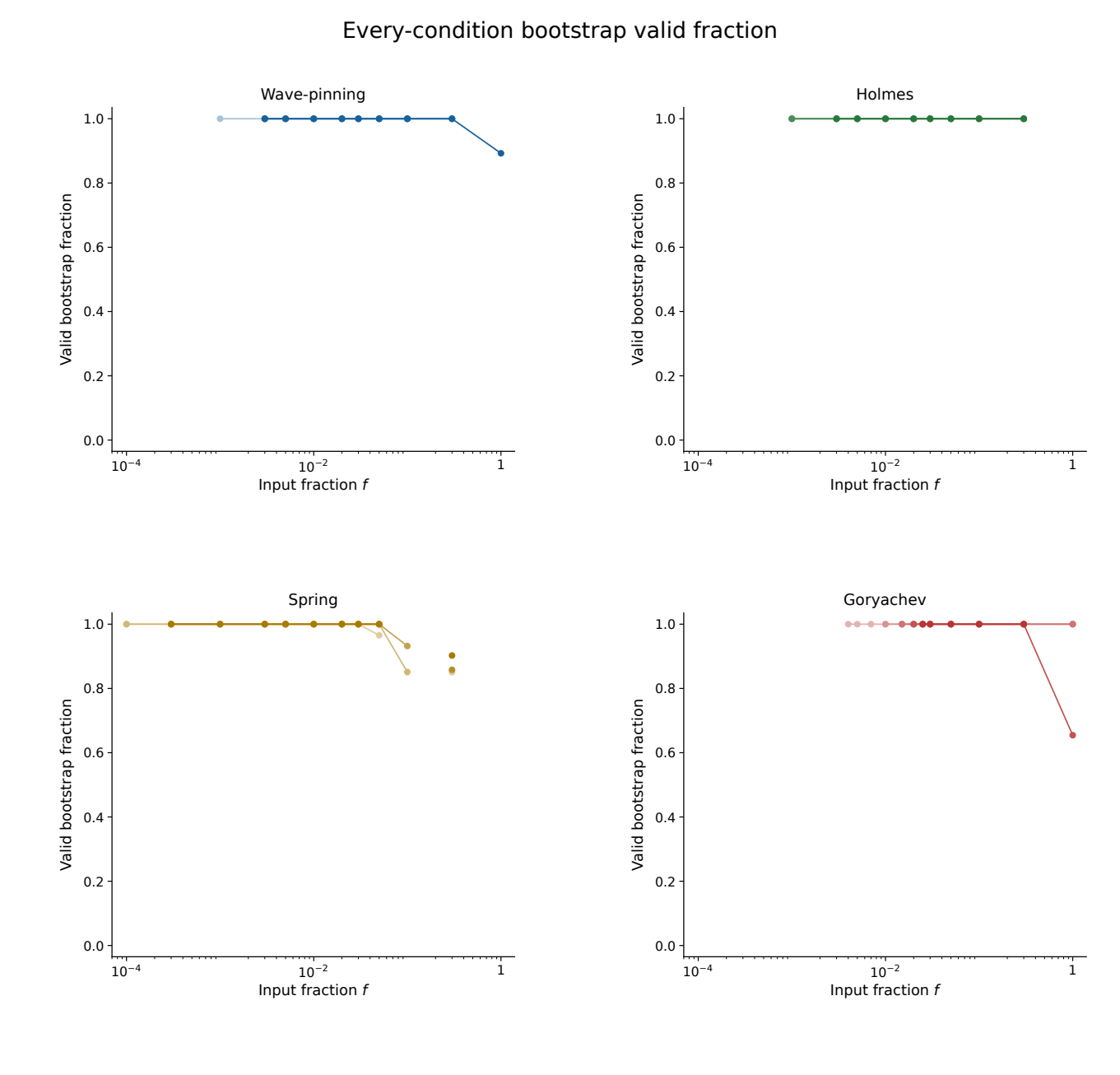


Dashed: hold 2; dotted: hold 10; dash-dot: MSD range 1/8; solid: MSD range 1/2. Primary hold 5, range 1/4.

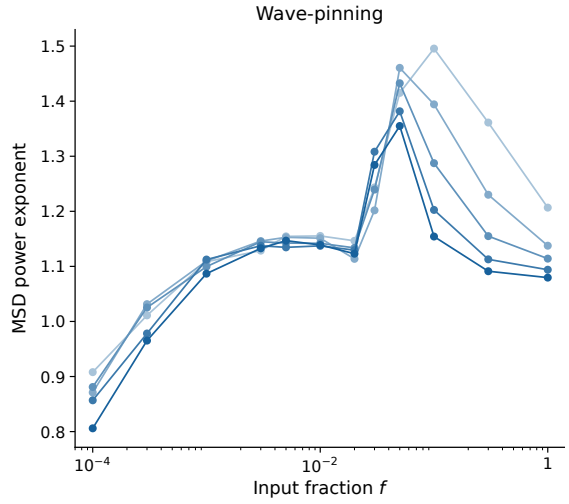


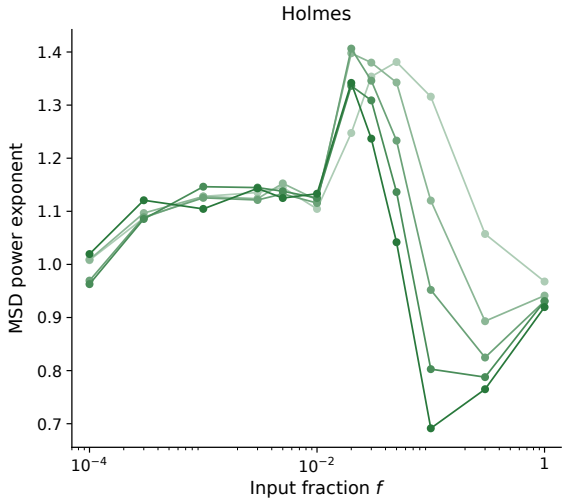
Dashed: hold 2; dotted: hold 10; dash-dot: MSD range 1/8; solid: MSD range 1/2. Primary hold 5, range 1/4.

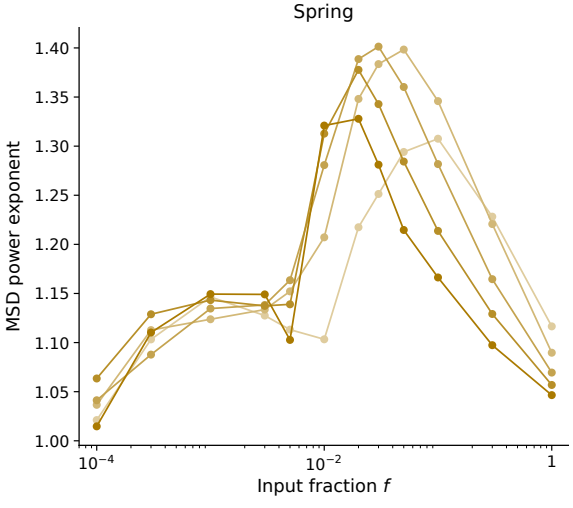

Every-condition valid fraction
Wave-pinning
Holmes
Spring
Goryachev
Valid heading fraction
Input fraction f
0.0
0.2
0.4
0.6
0.8
1.0
10^{-4}
10^{-2}
1

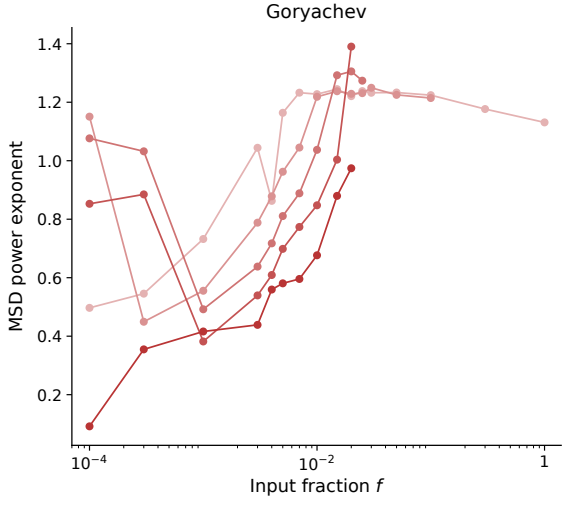

Every-condition bootstrap valid fraction
Wave-pinning
Holmes
Spring
Goryachev
Valid bootstrap fraction
Input fraction f
1.0
0.8
0.6
0.4
0.2
0.0
10^{-4}
10^{-2}
1

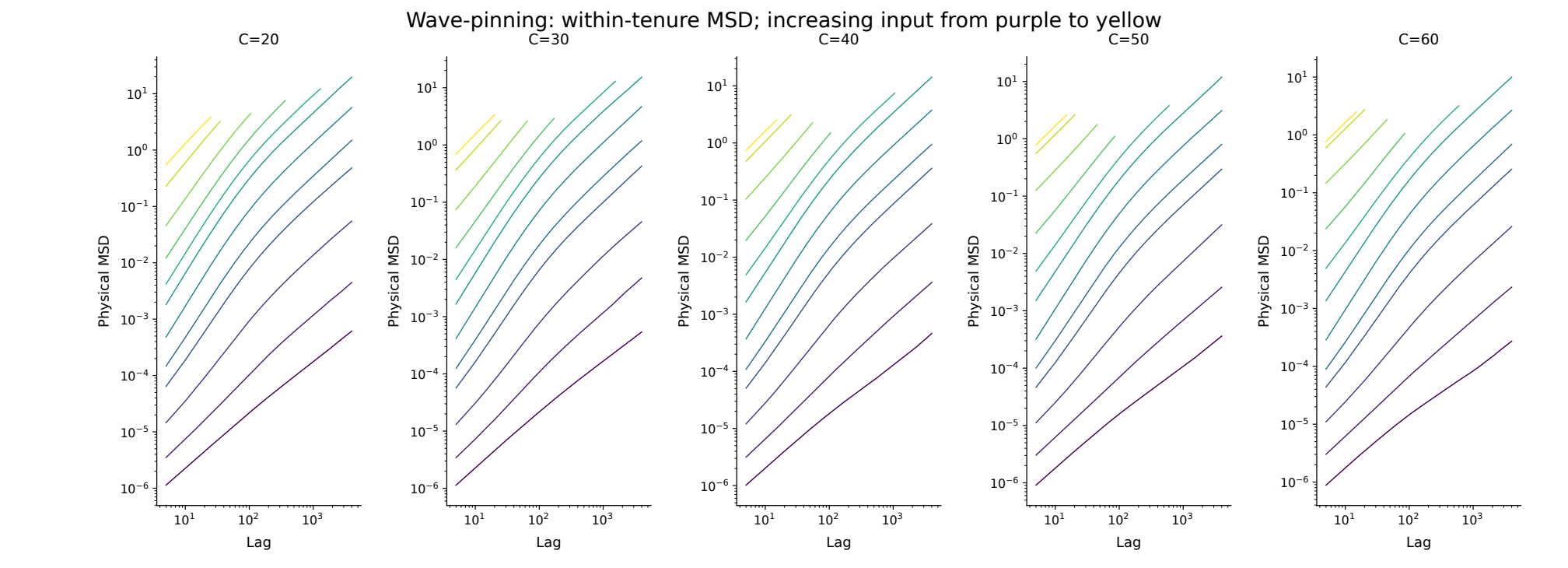
Within-tenure free MSD power exponent

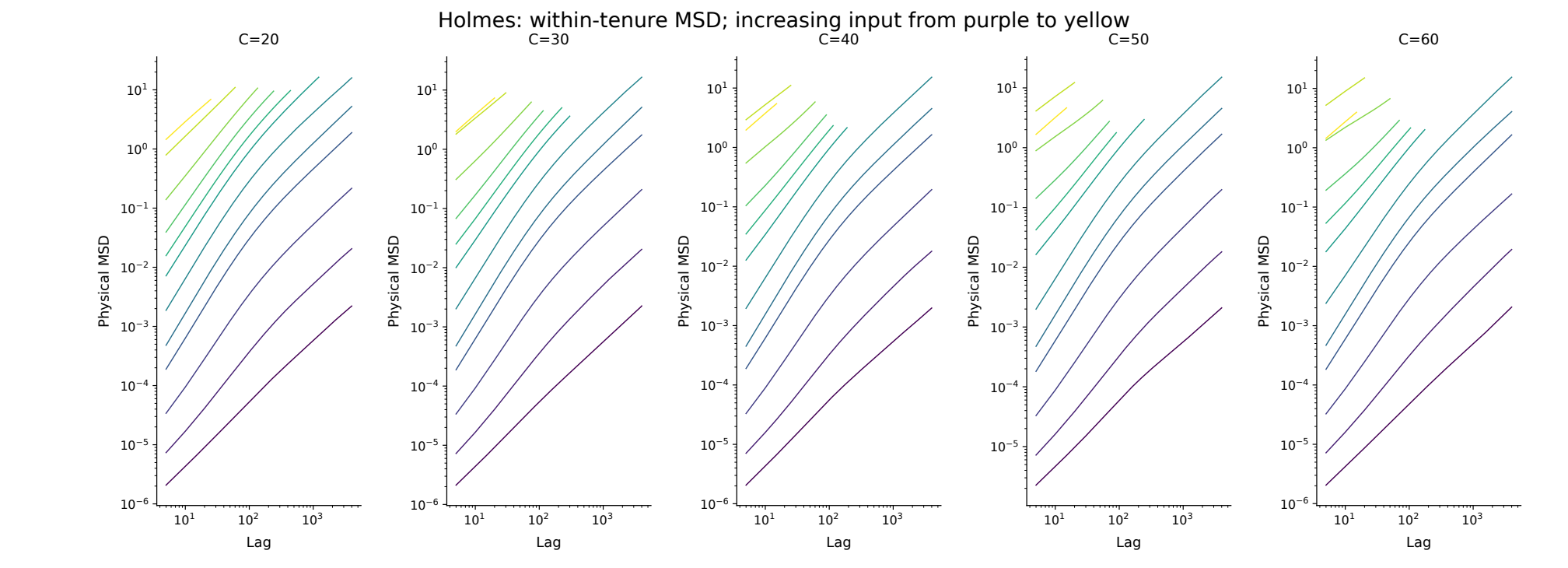


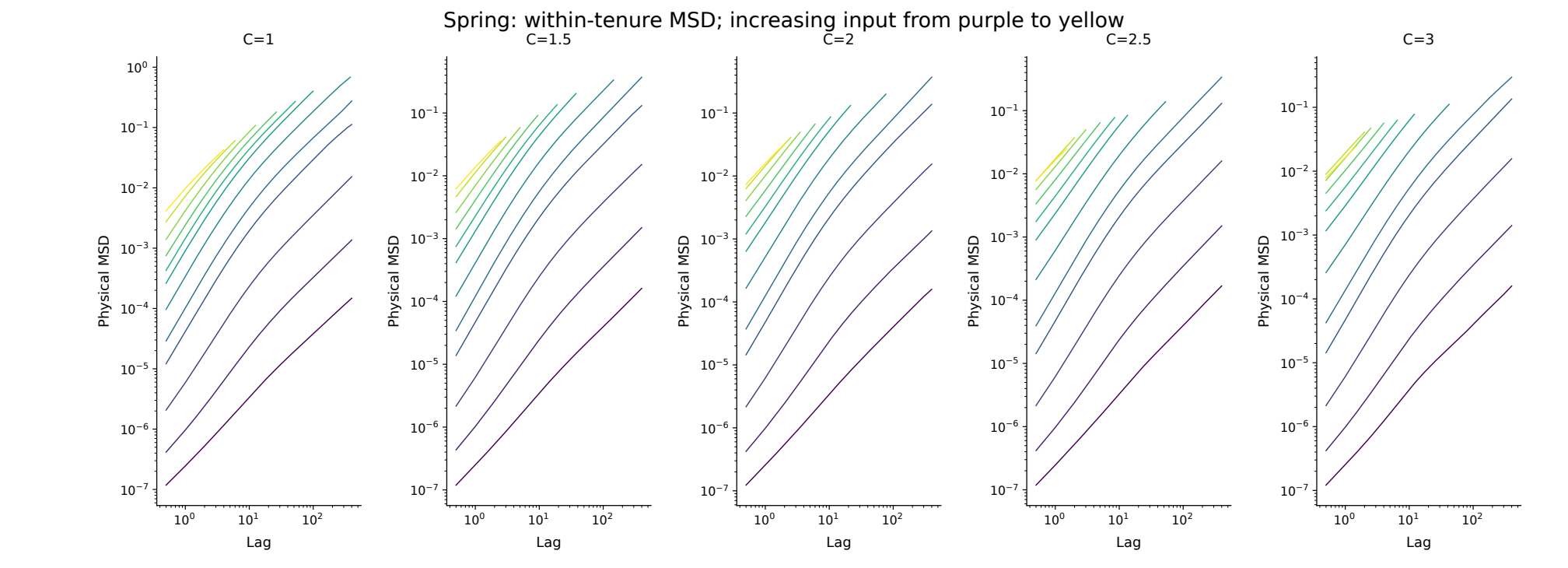


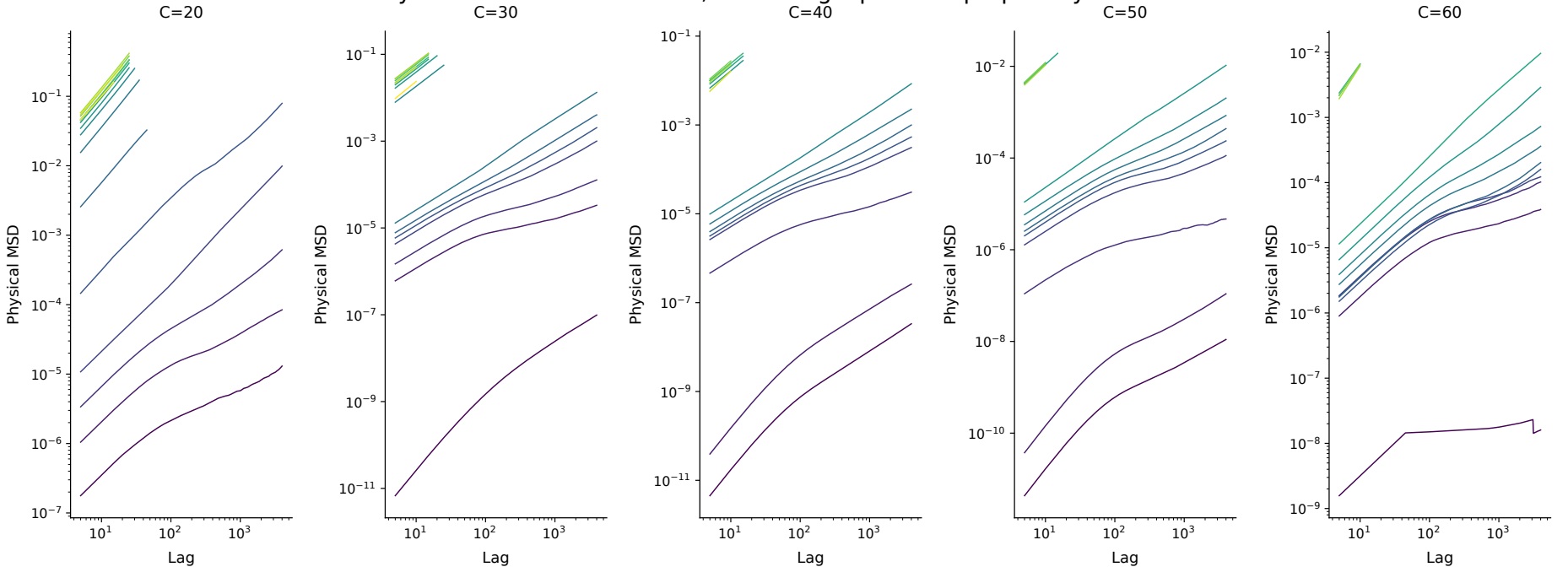


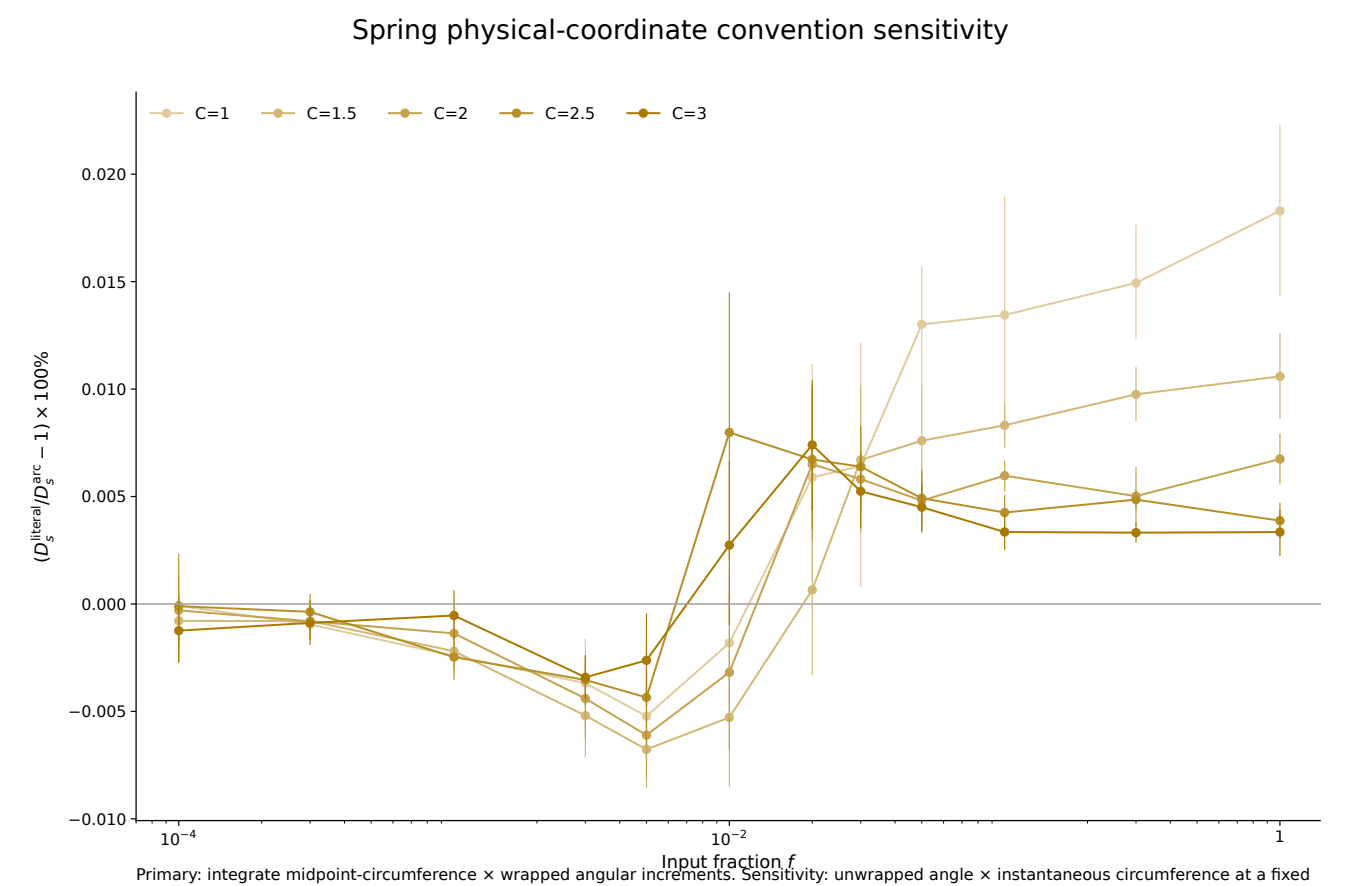



Wave-pinning: within-tenure MSD; increasing input from purple to yellow
C=20
C=30
C=40
C=50
C=60
Physical MSD
Lag


Holmes: within-tenure MSD; increasing input from purple to yellow
C=20
C=30
C=40
C=50
C=60
Physical MSD
Lag

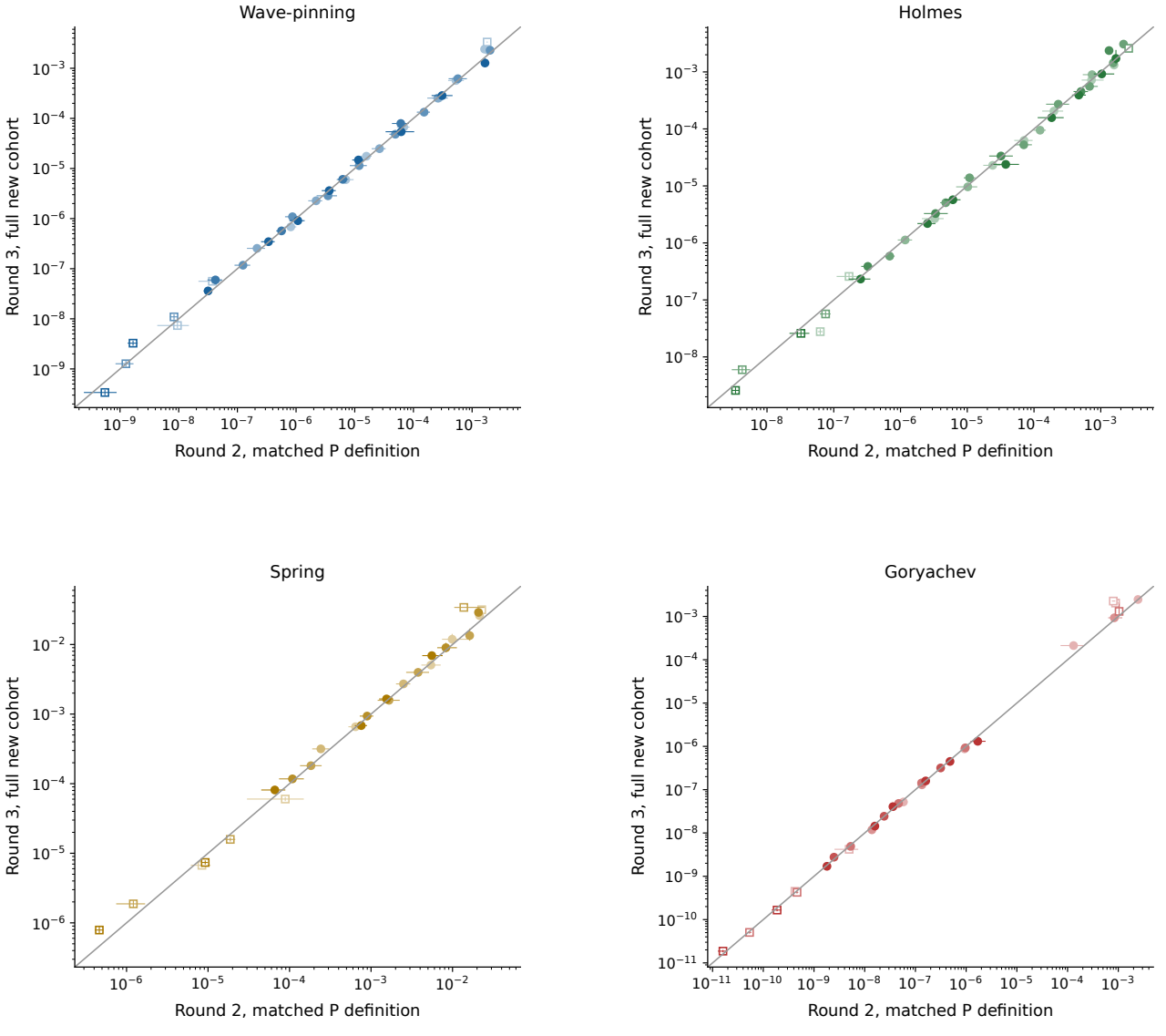

Spring: within-tenure MSD; increasing input from purple to yellow
C=1
C=1.5
C=2
C=2.5
C=3
Physical MSD
Lag


Goryachev: within-tenure MSD; increasing input from purple to yellow
C=20
C=30
C=40
C=50
C=60
Physical MSD
Lag



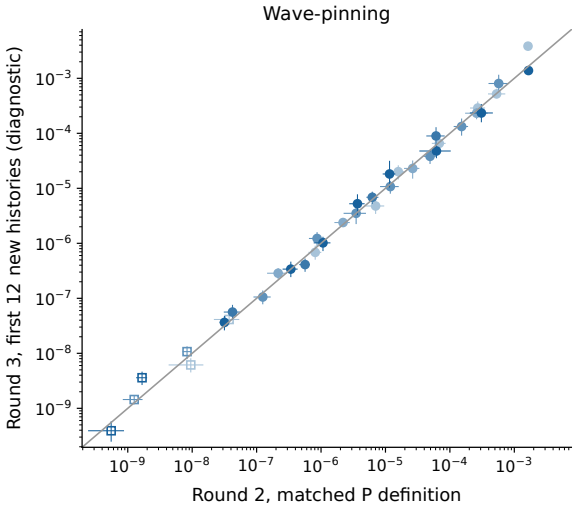
Primary: integrate midpoint-circumference × wrapped angular increments. Sensitivity: unwrapped angle × instantaneous circumference at a fixed coordinate origin. Whole-seed 95% intervals.

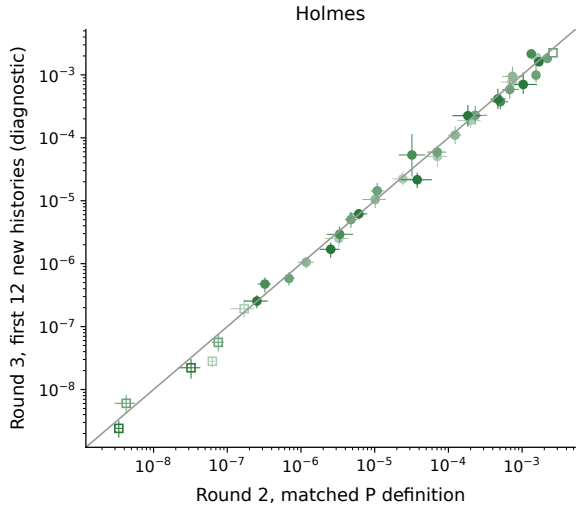
Round-2 versus round-3: Fixed-lag decay rate; full new cohort

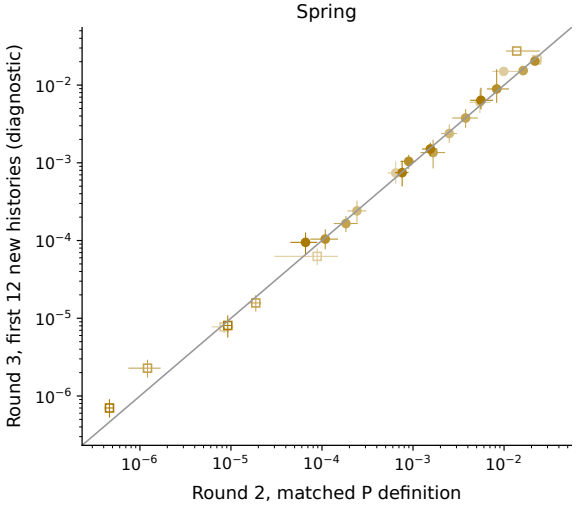


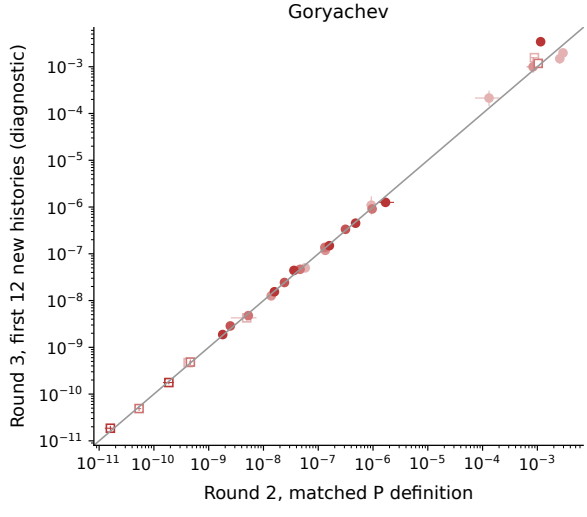
Filled circles: old ensemble has 12 histories. Open squares: old count differs or varies over the tolerance grid. Actual old/new counts and all intervals remain in CSV.

Round-2 versus round-3: Fixed-lag decay rate; first 12 new histories (diagnostic)



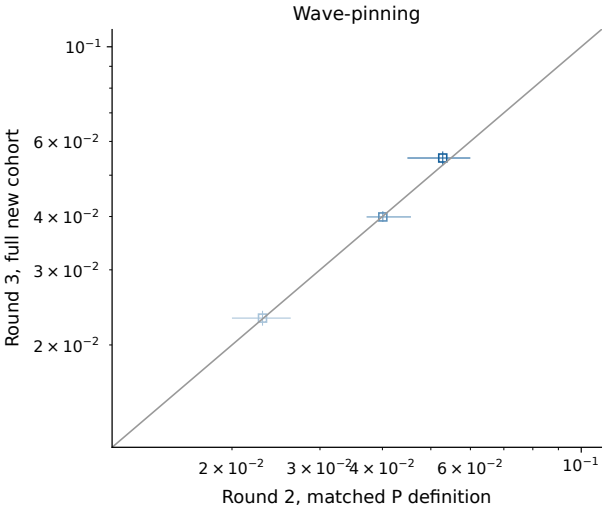


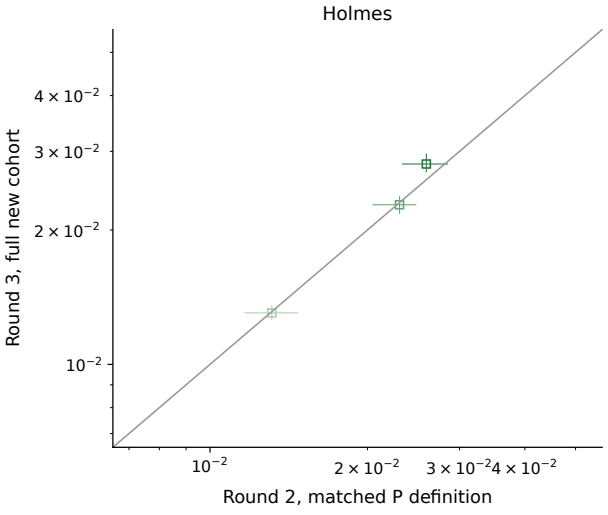


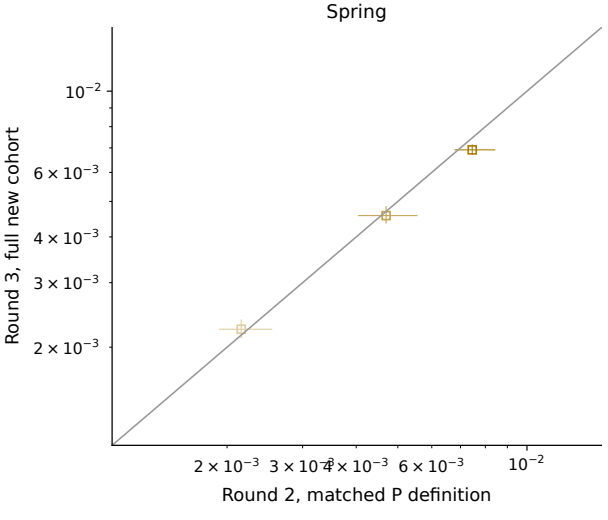


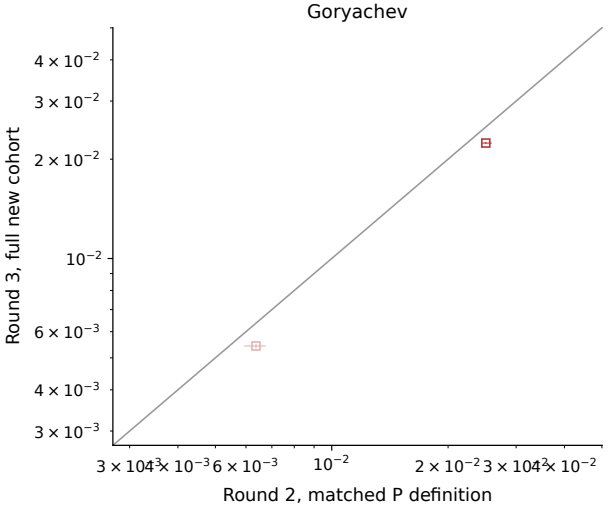
Filled circles: old ensemble has 12 histories. Open squares: old count differs or varies over the tolerance grid. Actual old/new counts and all intervals remain in CSV.

Round-2 versus round-3: Noise tolerance; full new cohort

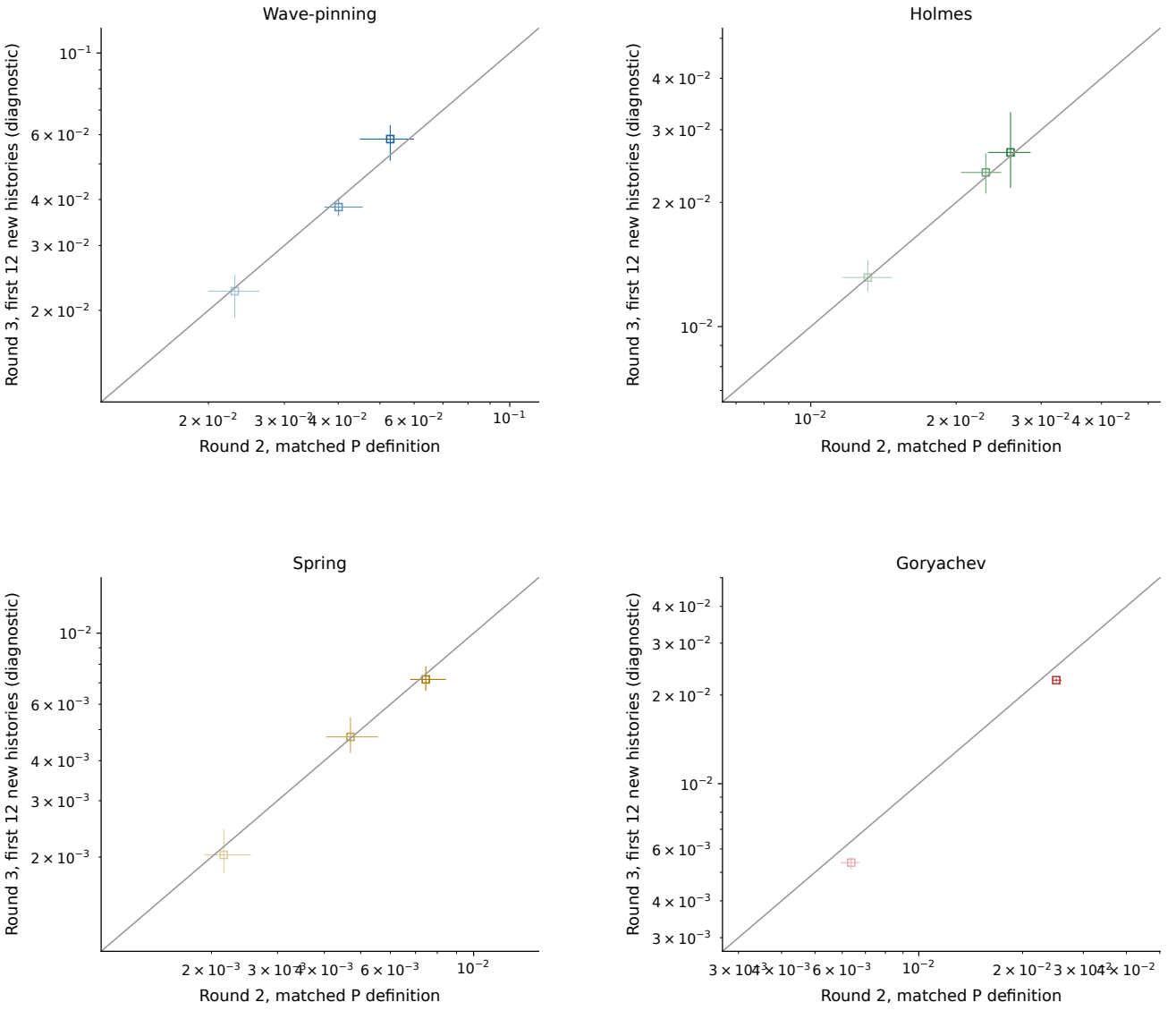








Filled circles: old ensemble has 12 histories. Open squares: old count differs or varies over the tolerance grid. Actual old/new counts and all intervals remain in CSV.

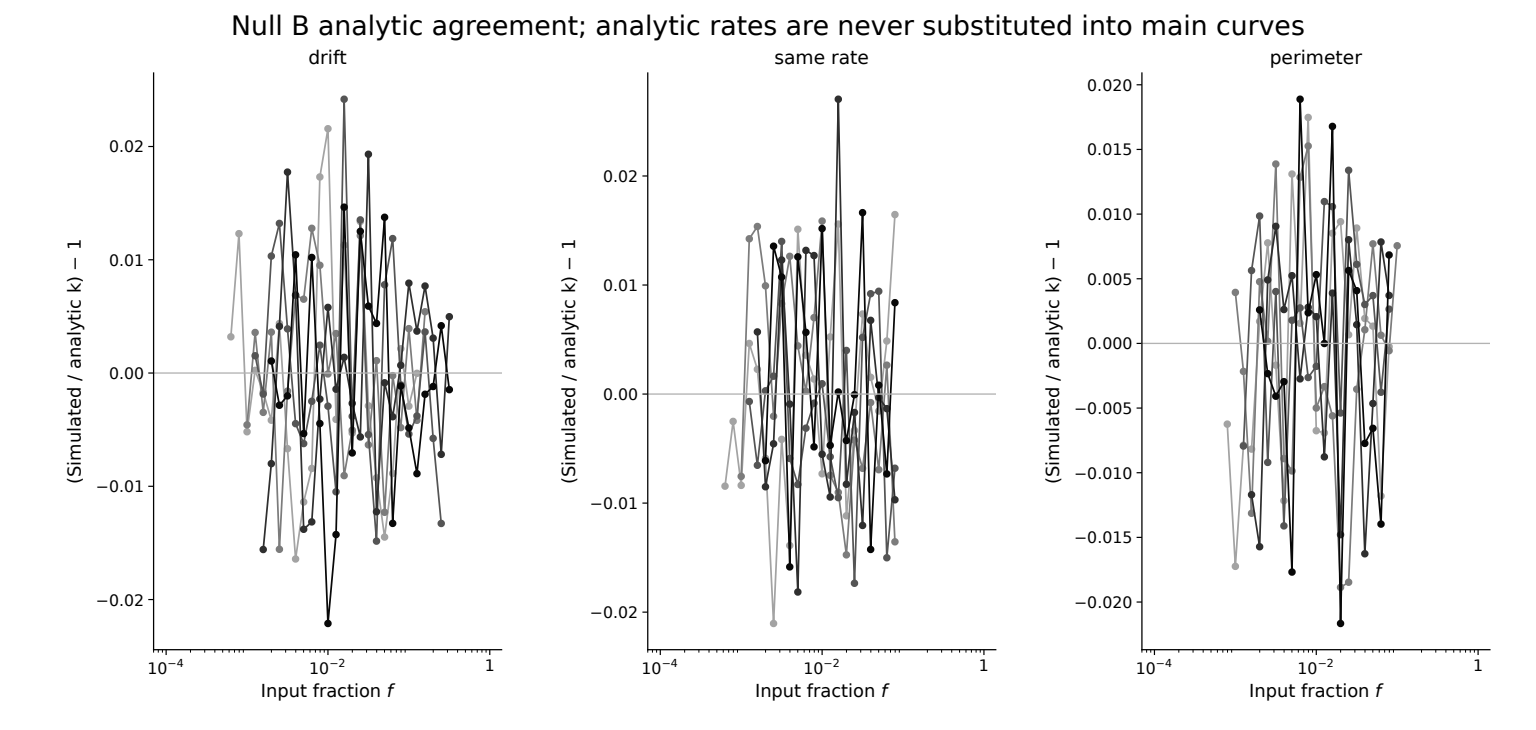

Round-2 versus round-3: Noise tolerance; first 12 new histories (diagnostic)
Wave-pinning
Holmes
Spring
Goryachev
Round 3, first 12 new histories (diagnostic)
Round 2, matched P definition
Filled circles: old ensemble has 12 histories. Open squares: old count differs or varies over the tolerance grid. Actual old/new counts and all intervals remain in CSV.

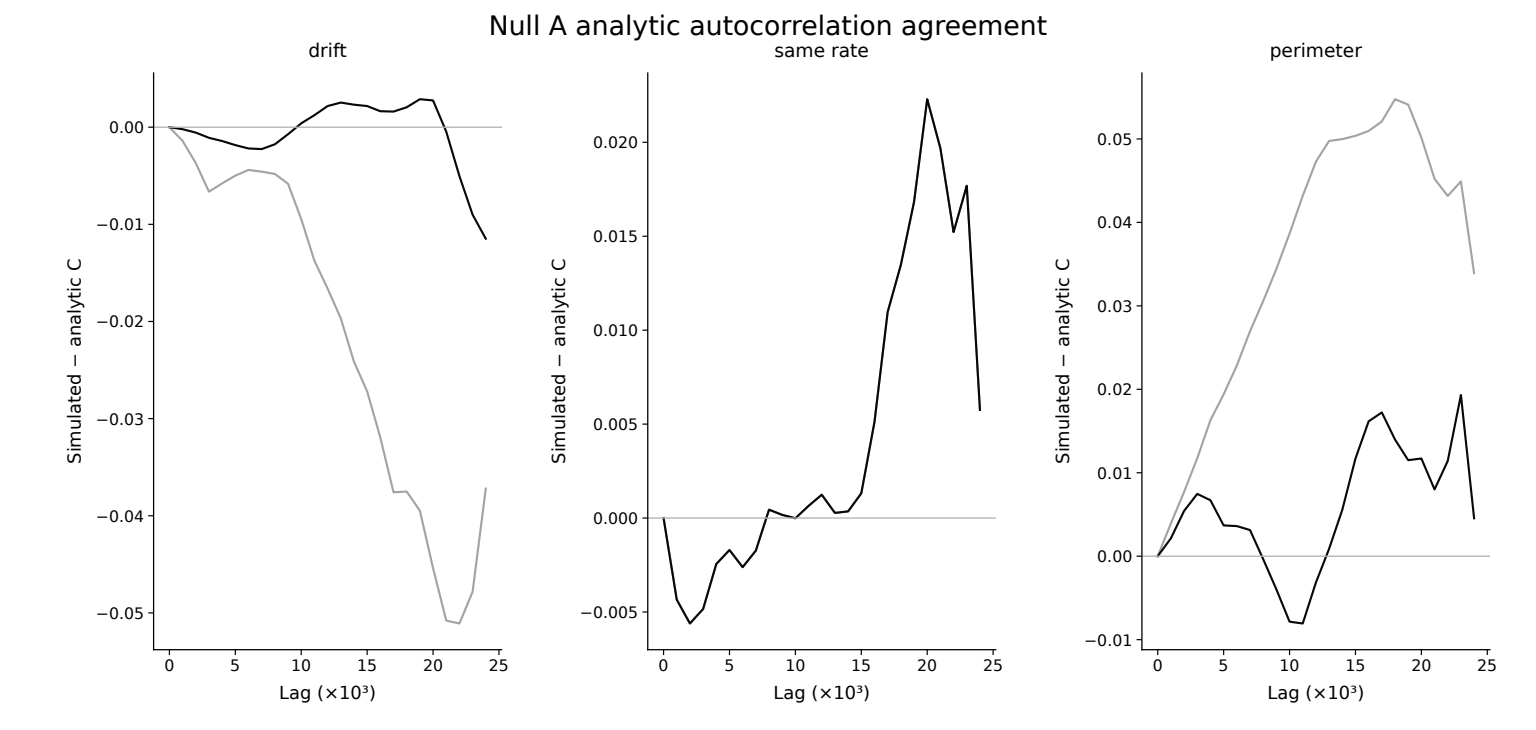
wave_pinning, C=20.0, f=0.001: n=384, horizon=32000.0
wave_pinning, C=20.0, f=0.003: n=384, horizon=32000.0
wave_pinning, C=20.0, f=0.005: n=384, horizon=32000.0
wave_pinning, C=20.0, f=0.01: n=384, horizon=32000.0
wave_pinning, C=20.0, f=0.3: n=384, horizon=32000.0
wave_pinning, C=20.0, f=1.0: n=192, horizon=32000.0
wave_pinning, C=30.0, f=0.003: n=384, horizon=32000.0
wave_pinning, C=30.0, f=0.005: n=384, horizon=32000.0
wave_pinning, C=30.0, f=0.01: n=384, horizon=32000.0
wave_pinning, C=30.0, f=0.02: n=288, horizon=32000.0
wave_pinning, C=30.0, f=0.3: n=192, horizon=32000.0
wave_pinning, C=40.0, f=0.003: n=384, horizon=32000.0
wave_pinning, C=40.0, f=0.005: n=384, horizon=32000.0
wave_pinning, C=40.0, f=0.01: n=384, horizon=32000.0
wave_pinning, C=40.0, f=0.02: n=384, horizon=32000.0
wave_pinning, C=40.0, f=0.03: n=192, horizon=32000.0
wave_pinning, C=50.0, f=0.003: n=384, horizon=32000.0
wave_pinning, C=50.0, f=0.005: n=384, horizon=32000.0
wave_pinning, C=50.0, f=0.01: n=384, horizon=32000.0
wave_pinning, C=50.0, f=0.02: n=384, horizon=32000.0
wave_pinning, C=50.0, f=0.03: n=288, horizon=32000.0
wave_pinning, C=60.0, f=0.003: n=384, horizon=32000.0
wave_pinning, C=60.0, f=0.005: n=384, horizon=32000.0
wave_pinning, C=60.0, f=0.01: n=384, horizon=32000.0
wave_pinning, C=60.0, f=0.02: n=384, horizon=32000.0
wave_pinning, C=60.0, f=0.03: n=384, horizon=32000.0
wave_pinning, C=60.0, f=0.05: n=192, horizon=32000.0
wave_pinning, C=60.0, f=1.0: n=384, horizon=32000.0
holmes_model3, C=20.0, f=0.001: n=384, horizon=32000.0
holmes_model3, C=20.0, f=0.003: n=384, horizon=32000.0
holmes_model3, C=20.0, f=0.005: n=384, horizon=32000.0
holmes_model3, C=20.0, f=0.3: n=192, horizon=32000.0
holmes_model3, C=20.0, f=1.0: n=192, horizon=32000.0
holmes_model3, C=30.0, f=0.001: n=384, horizon=32000.0
holmes_model3, C=30.0, f=0.003: n=384, horizon=32000.0

holmes_model3, C=30.0, f=0.005: n=384, horizon=32000.0
holmes_model3, C=30.0, f=0.01: n=288, horizon=32000.0
holmes_model3, C=30.0, f=1.0: n=384, horizon=32000.0
holmes_model3, C=40.0, f=0.001: n=384, horizon=32000.0
holmes_model3, C=40.0, f=0.003: n=384, horizon=32000.0
holmes_model3, C=40.0, f=0.005: n=384, horizon=32000.0
holmes_model3, C=40.0, f=0.01: n=384, horizon=32000.0
holmes_model3, C=50.0, f=0.001: n=384, horizon=32000.0
holmes_model3, C=50.0, f=0.003: n=384, horizon=32000.0
holmes_model3, C=50.0, f=0.005: n=384, horizon=32000.0
holmes_model3, C=50.0, f=0.01: n=384, horizon=32000.0
holmes_model3, C=50.0, f=0.02: n=192, horizon=32000.0
holmes_model3, C=50.0, f=0.3: n=192, horizon=32000.0
holmes_model3, C=60.0, f=0.003: n=384, horizon=32000.0
holmes_model3, C=60.0, f=0.005: n=384, horizon=32000.0
holmes_model3, C=60.0, f=0.01: n=384, horizon=32000.0
holmes_model3, C=60.0, f=0.02: n=192, horizon=32000.0
spring_ring, C=1.0, f=0.0001: n=384, horizon=3200.0
spring_ring, C=1.0, f=0.0003: n=384, horizon=3200.0
spring_ring, C=1.0, f=0.001: n=384, horizon=3200.0
spring_ring, C=1.0, f=0.05: n=384, horizon=3200.0
spring_ring, C=1.0, f=0.3: n=384, horizon=3200.0
spring_ring, C=1.5, f=0.0001: n=384, horizon=3200.0
spring_ring, C=1.5, f=0.0003: n=384, horizon=3200.0
spring_ring, C=1.5, f=0.001: n=384, horizon=3200.0
spring_ring, C=1.5, f=0.1: n=384, horizon=3200.0
spring_ring, C=1.5, f=0.3: n=192, horizon=3200.0
spring_ring, C=2.0, f=0.0003: n=384, horizon=3200.0
spring_ring, C=2.0, f=0.001: n=384, horizon=3200.0
spring_ring, C=2.0, f=0.05: n=192, horizon=3200.0
spring_ring, C=2.0, f=0.1: n=384, horizon=3200.0
spring_ring, C=2.5, f=0.0003: n=384, horizon=3200.0
spring_ring, C=2.5, f=0.001: n=384, horizon=3200.0
spring_ring, C=2.5, f=0.003: n=288, horizon=3200.0
spring_ring, C=2.5, f=0.1: n=192, horizon=3200.0

Extended horizons and conditions above 96 seeds (71–84)

spring_ring, C=2.5, f=0.3: n=384, horizon=3200.0
spring_ring, C=3.0, f=0.0003: n=384, horizon=3200.0
spring_ring, C=3.0, f=0.001: n=384, horizon=3200.0
spring_ring, C=3.0, f=0.003: n=384, horizon=3200.0
spring_ring, C=3.0, f=0.3: n=384, horizon=3200.0
goryachev, C=20.0, f=0.004: n=384, horizon=32000.0
goryachev, C=20.0, f=0.005: n=192, horizon=32000.0
goryachev, C=30.0, f=1.0: n=192, horizon=32000.0
goryachev, C=40.0, f=1.0: n=384, horizon=32000.0
goryachev, C=50.0, f=0.015: n=192, horizon=32000.0
goryachev, C=50.0, f=0.3: n=384, horizon=32000.0
goryachev, C=50.0, f=1.0: n=384, horizon=32000.0
goryachev, C=60.0, f=0.1: n=192, horizon=32000.0
goryachev, C=60.0, f=0.3: n=384, horizon=32000.0


Null B analytic agreement; analytic rates are never substituted into main curves
drift
same rate
perimeter
(Simulated / analytic k) − 1
Input fraction f
0.02
0.01
0.00
−0.01
−0.02
0.015
0.010
0.005
−0.005
−0.010
−0.015
−0.020
10^{-4}
10^{-2}
1


Null A analytic autocorrelation agreement
drift
same rate
perimeter
Simulated − analytic C
Lag (×10³)
0.00
−0.01
−0.02
−0.03
−0.04
−0.05
0.020
0.015
0.010
0.005
0.000
−0.005
0.05
0.04
0.03
0.02
0.01
0.00
−0.01
0
5
10
15
20
25

Every local neighborhood: circle resolved, cross unavailable; reasons in compact CSV

